Fiscal Year:	FY 2018	**Task Last Updated:**	FY 03/12/2018
PI Name:	Weber, Richard Ph.D.		
Project Title:	Microgravity Investigation of Thermophysical Properties of Supercooled Molten Metal Oxides		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
Organization Name:	Materials Development Inc.		
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PI Web Page:			
City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/21/2017	**End Date:**	05/20/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX17AH73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed research will implement microgravity based experiments using the JAXA (Japanese Aerospace Exploration Agency) flight Electrostatic Levitation Furnace (ELF) to achieve precise measurements of the thermophysical properties of supercooled (also called undercooled) molten metal oxides. The work will include both ground based and flight experiments on liquids formed from aluminum, calcium, silicon, and rare earth element oxide mixtures. These liquids are precursors to high value-added glass materials that are used in photonics, lasers, optical communications, and imaging applications. The outcome of the proposed international research collaboration will be to: (i) Measure technologically important thermophysical properties of normal and supercooled liquid metal oxides of interest in connection with production of photonic and optical device materials, (ii) Develop thermophysical property datasets that complement existing atomic structure data on the materials under investigation. Use the resulting data to increase understanding of the nature of fragile liquids and the glass transition, and (iii) Disseminate the new data via publication, through NASA's Physical Sciences Informatics System and the Materials Genome Initiative to help accelerate the development of new products. We will investigate the behavior of liquid phase metal oxides as they transform to glass in quiescent, diffusion-controlled (i.e., without fluid motion) conditions. This work is needed to optimize photonic and optical device materials to obtain uniform properties and stable performance in applications. Overall the work will advance the broader goals of optimizing computational modeling of materials processes and developing superior materials by design. The flight ELF can achieve pristine molten and supercooled samples. Such samples can be studied using non-contact measurements of temperature, volume, surface tension, and viscosity. Spherical samples ca. 2 mm in diameter suitable for experiments in the flight ELF will be made by laser hearth melting. Prior ground-based work demonstrated that the sample materials can be: (i) levitated using electrostatic methods, (ii) heated using laser beams, and (iii) melted without significant evaporation. Proposed ground-based work will push the limits of these measurements and help to optimize the methods for analysis for the flight data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work relates to development, characterization, and optimization of functional glasses and glass ceramics. These materials are high value add products that are using in telecommunications, displays (TVs and phones), medical lasers, and security sensor equipment. The project work includes outreach to undergraduate and high school students.
Task Progress:	Work is in progress to establish requirements and samples synthesis for experiments planned for the International Space Station (ISS). During the first year conditions for heating samples were established and ground based electrostatic levitation experiments were performed at NASA Marshall Space Flight Center. The Principal Investigator (PI) visited the Japanese collaborators to discuss experiments with the ISS levitation instrument that is operated by the Japanese space agency (JAXA). Additional data analysis was performed and one technical article was published. A detailed report was submitted to NASA.
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Kohara S, Ohara K, Ishikawa T, Tamaru H, Weber R. "Investigation of structure and dynamics in disordered materials using containerless techniques with in-situ quantum beam and thermophysical property measurements." Quantum Beam Science. 2018 Mar;2(1):5. https://doi.org/10.3390/qubs2010005 , Mar-2018